Block Box

DEVELOPING A SECURE LOCKER WITH PARCEL VERIFICATION FOR BLOCKCHAIN INTEGRATION

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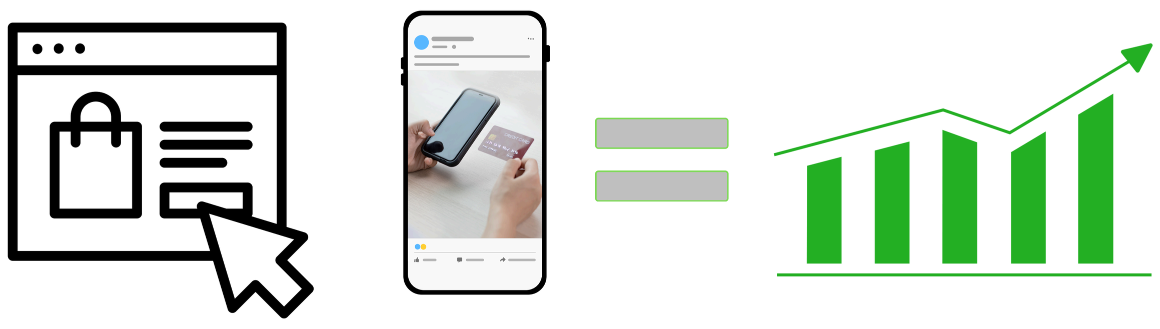
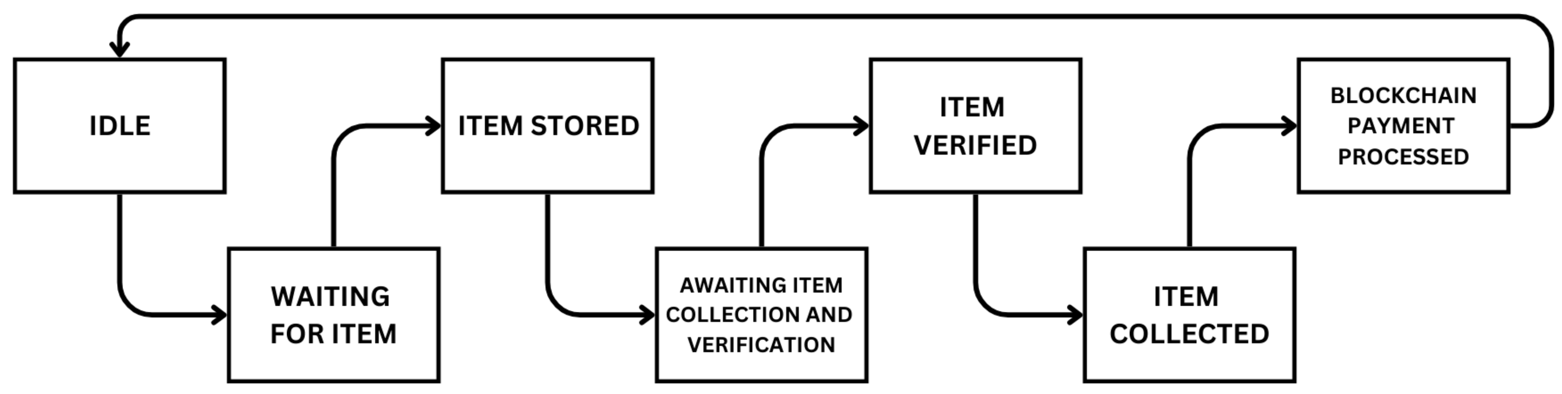
1 BACKGROUND



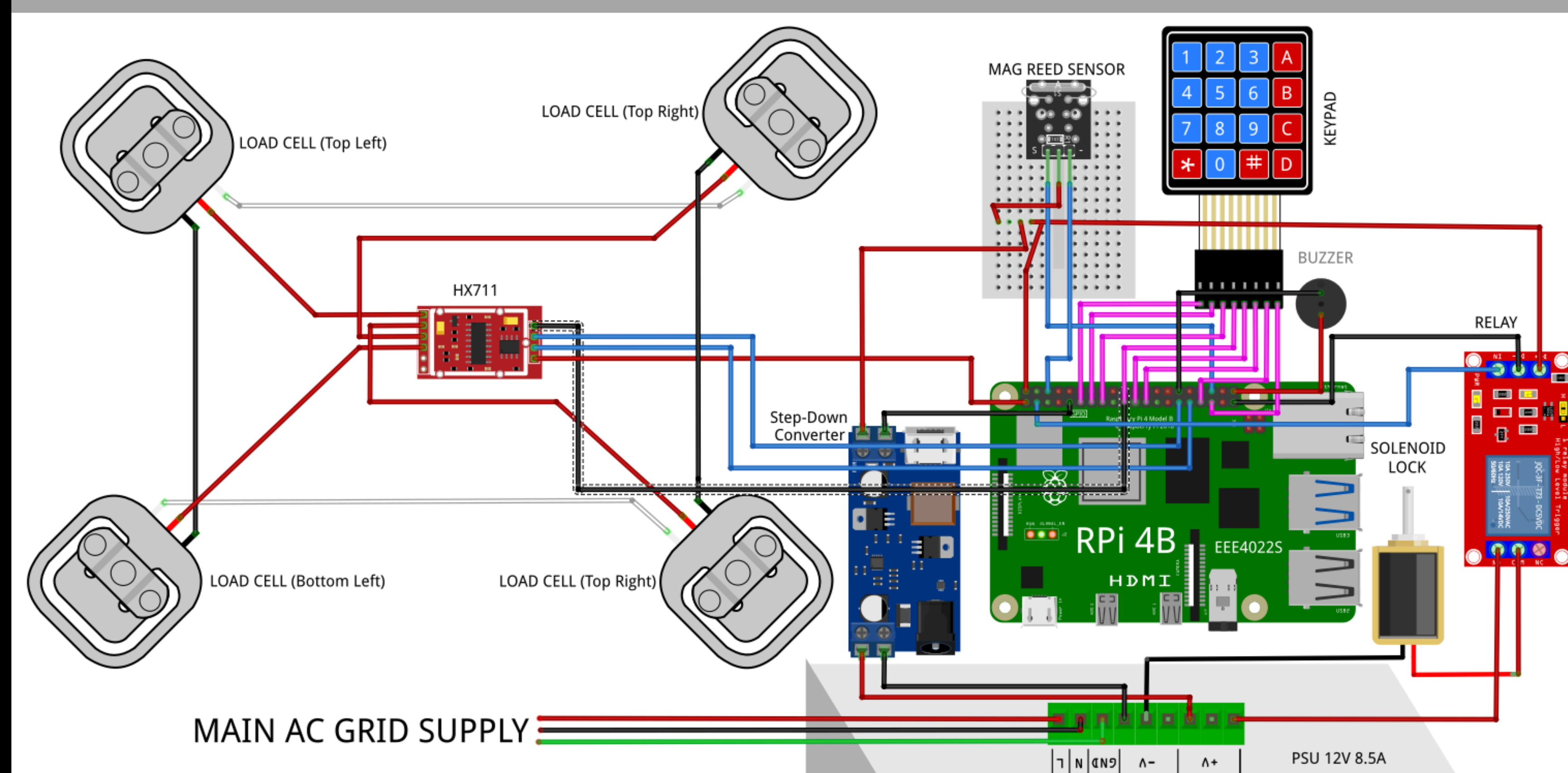
E-commerce growth has led to an increase in peer-to-peer transactions that require secure and reliable last-mile delivery (LMD) solutions. Traditional LMD systems pose challenges related to asymmetric information between buyers and sellers.

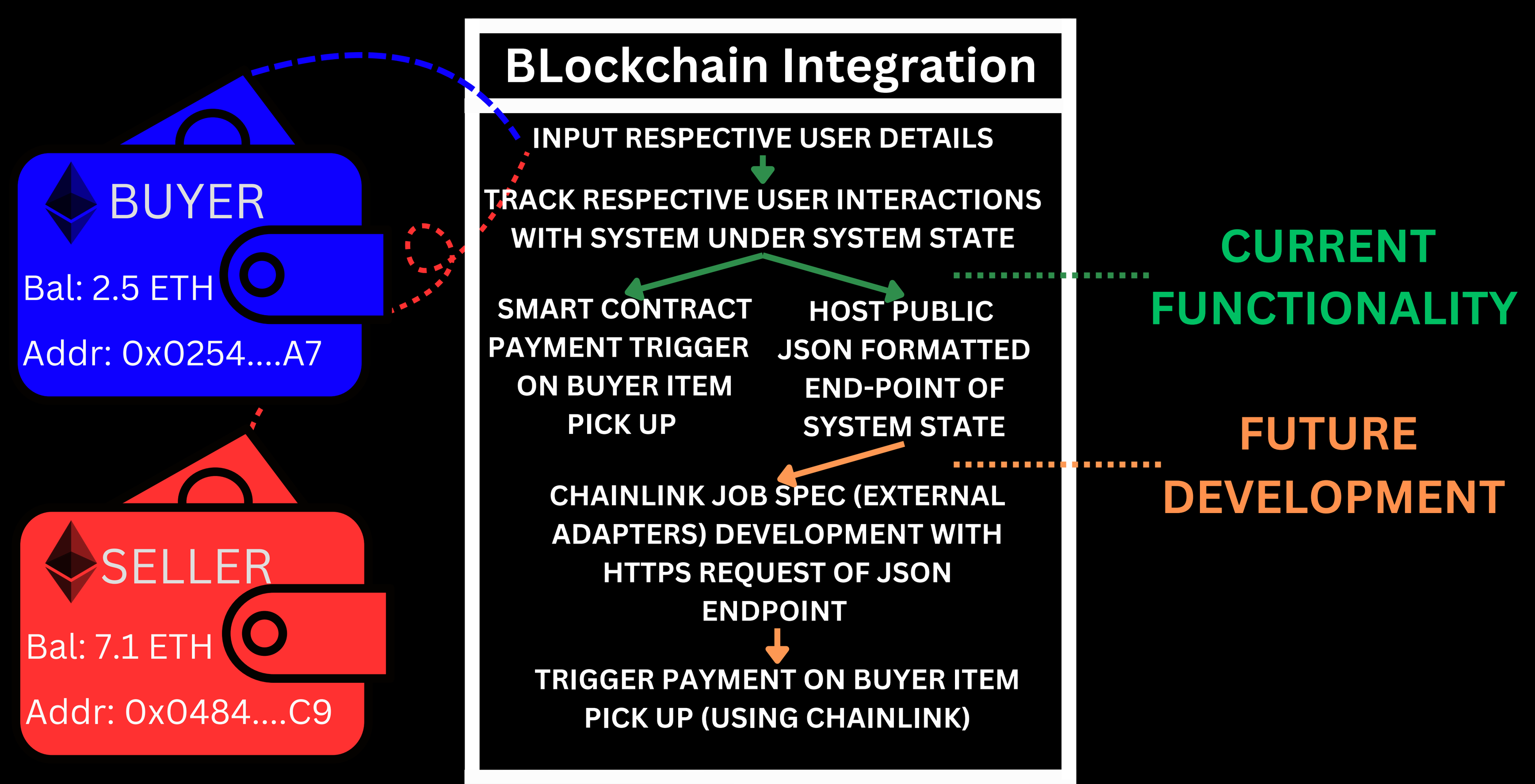
2 APPROACH

The seller typically has more information about the transaction (e.g. condition of the product) when payment takes place. BLock Box solves the problem with the following system states:



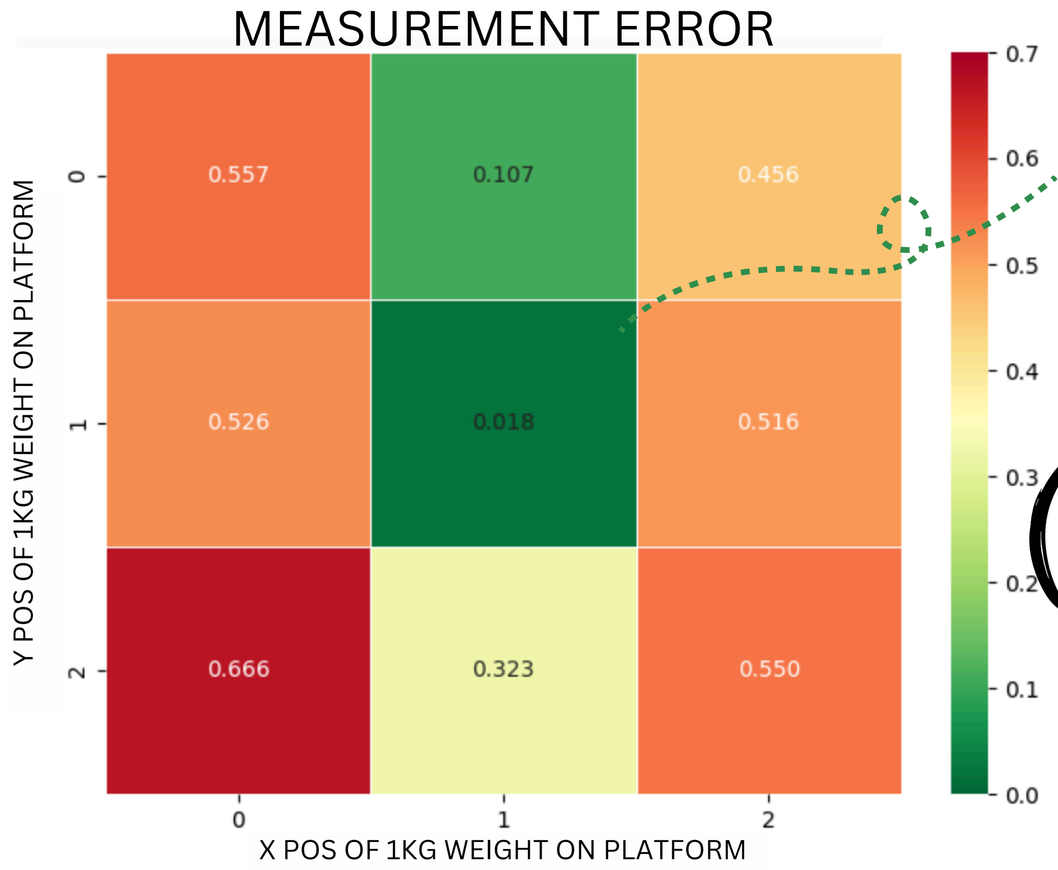
3 IMPLEMENTATION



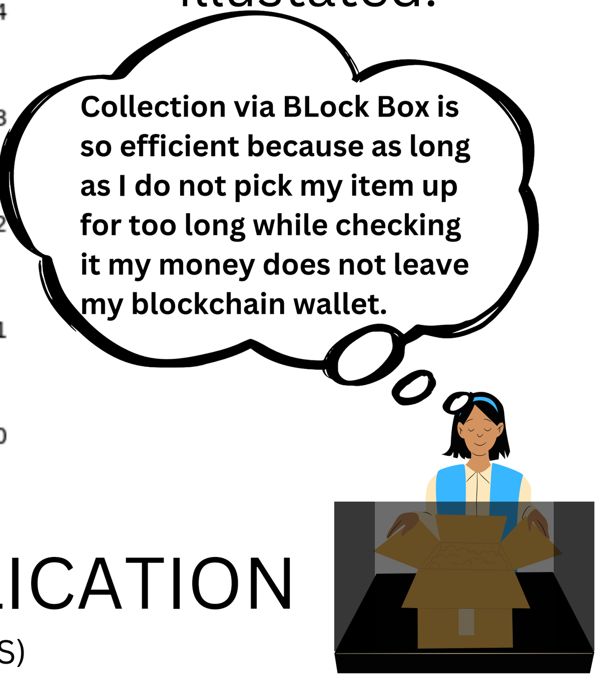


4 RESULTS

SENSOR PLATFORM ACCURACY
MEASUREMENT ERROR

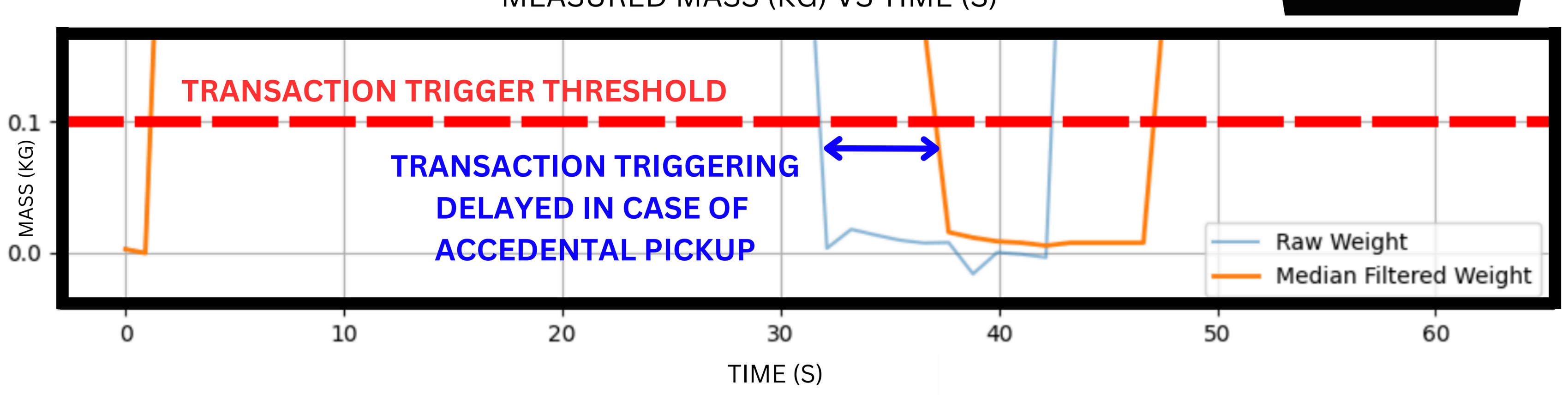


*Scale is the most accurate at the center as off-center error can be quite large as illustated.



MEDIAN FILTER APPLICATION

MEASURED MASS (KG) VS TIME (S)



5 CONCLUSION

A secure parcel locker system with automated verification using weight sensors was developed. Upon successful verification, the system uses a smart contract payment system to automate the payment process, therefore, balancing the scale.

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